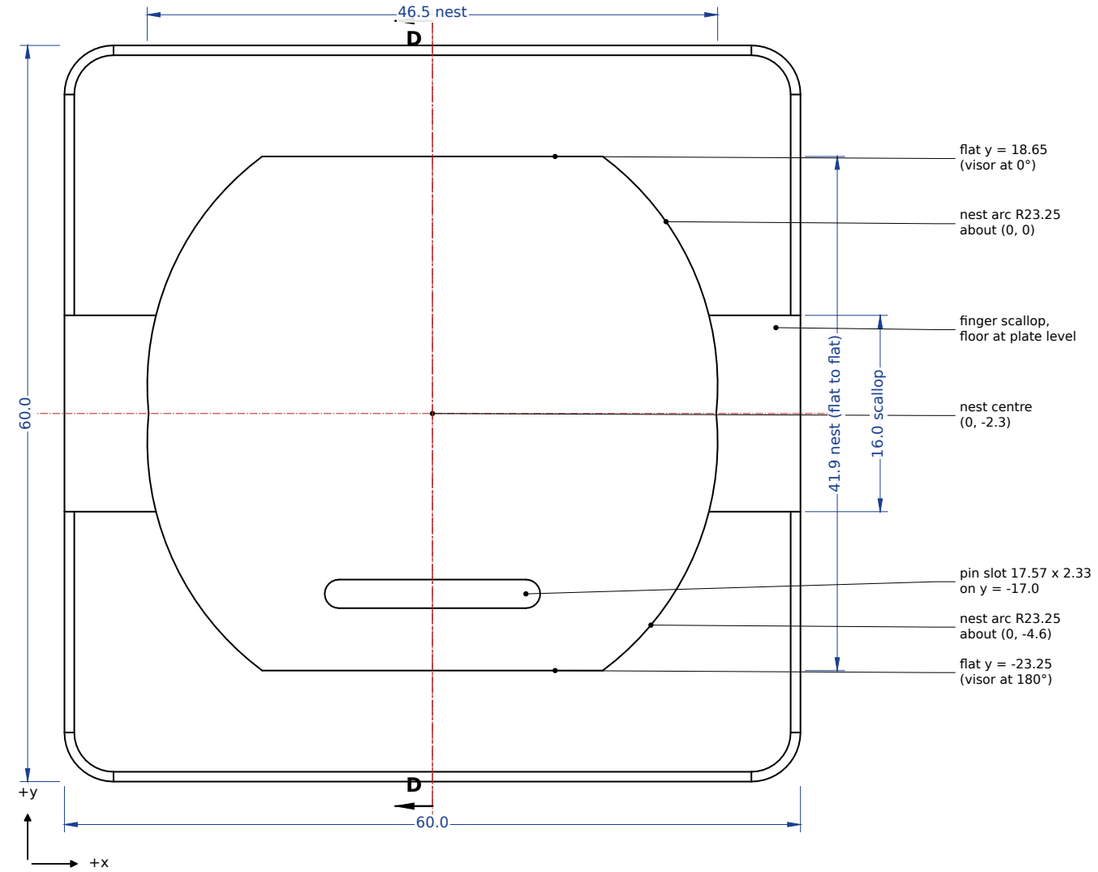
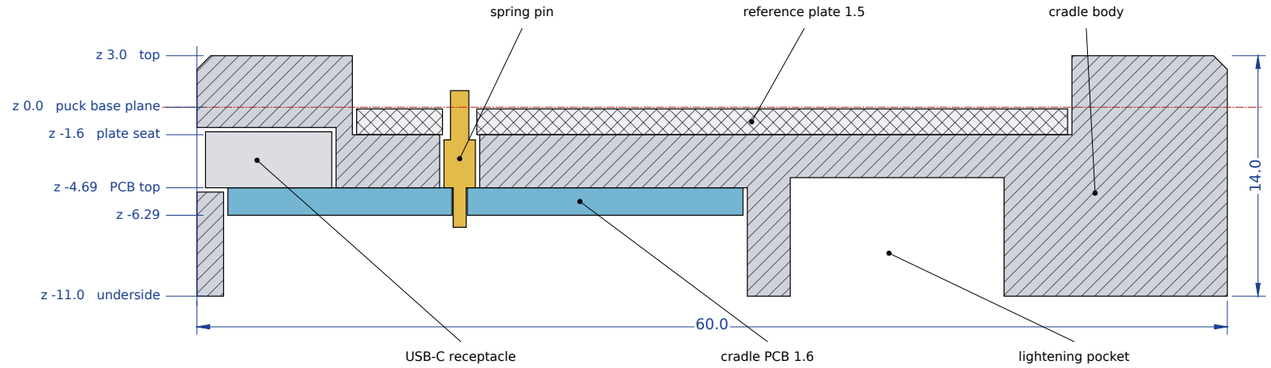
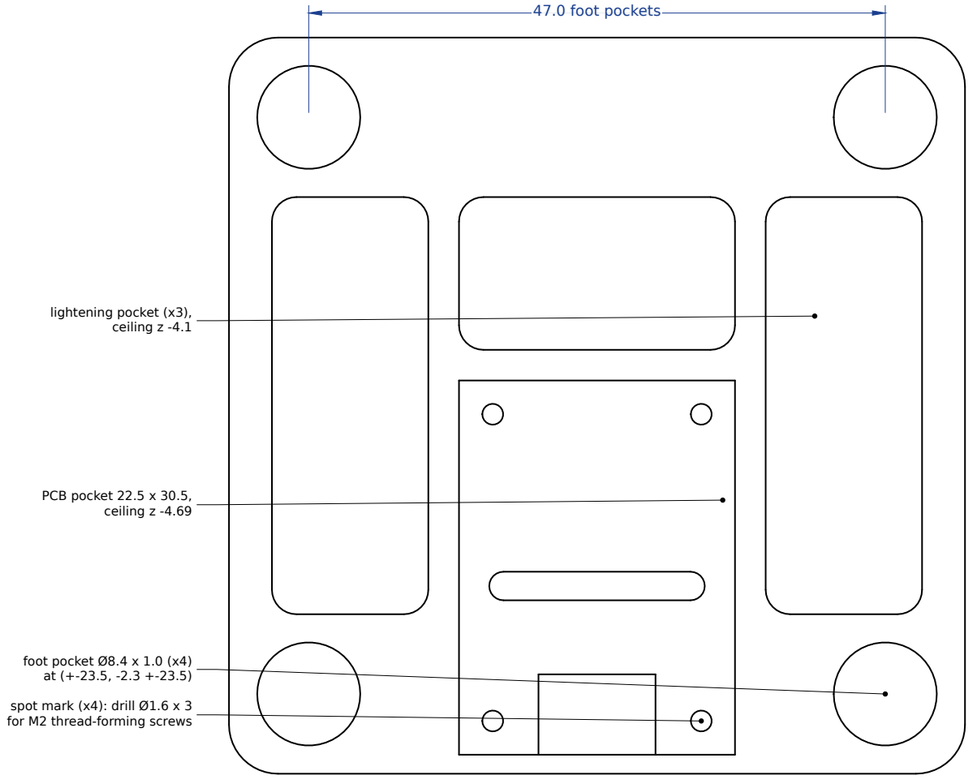
## TOP VIEW  2:1




## SECTION D-D  (x = 0) with plate, PCB, receptacle and centre pin  2.8:1




## VIEW FROM BELOW  2:1  (+x to the LEFT)



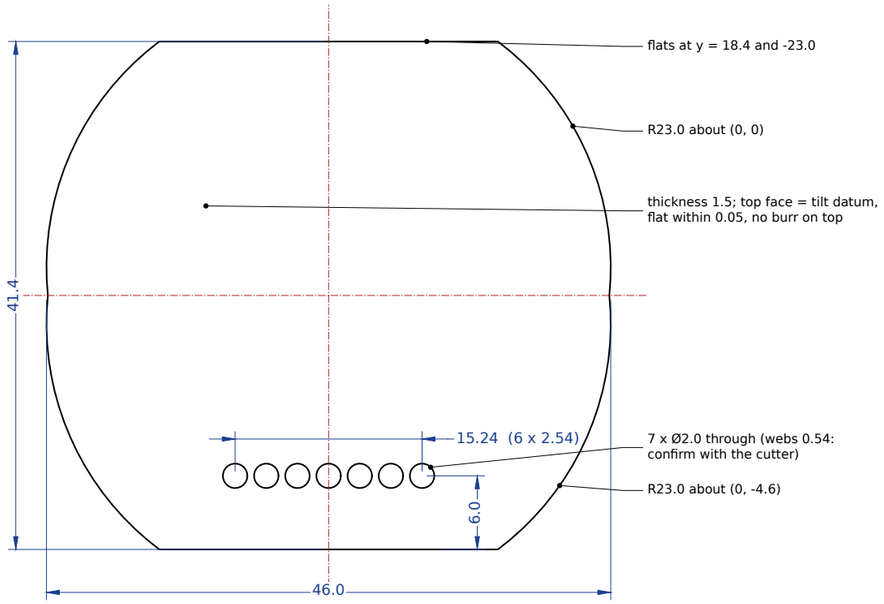

### NOTES

1. Printed body 60 x 60 x 14 mm. The nest is the union of the puck outline (+0.25) at 0° and turned 180° about the nest centre: it accepts the marker in exactly these two positions; its centre moves 4.6 mm between them (deviation D15).
2. Seat for the stainless reference plate at z = -1.6; the three ball feet rest on the plate top (z = -0.1). Bond the plate with a thin epoxy film on three pads only, then check flatness - the plate, not the print, is the tilt reference.
3. Cradle PCB top at z = -4.69 so that each spring pin is at its recommended working height (5.82) on a pad 1.13 above the puck base plane.
4. OPEN (D17): the seven pins push the 11 g marker up with about 4 N. No hold-down is modelled; one is required before the cradle can charge reliably or serve as the flip-calibration seat. At 180° the pins bear on the solid base plate (no pad, no opening).
5. Pin and receptacle envelopes are PROVISIONAL. Sheet 2: reference plate, pin row section, 0° / 180° positions.

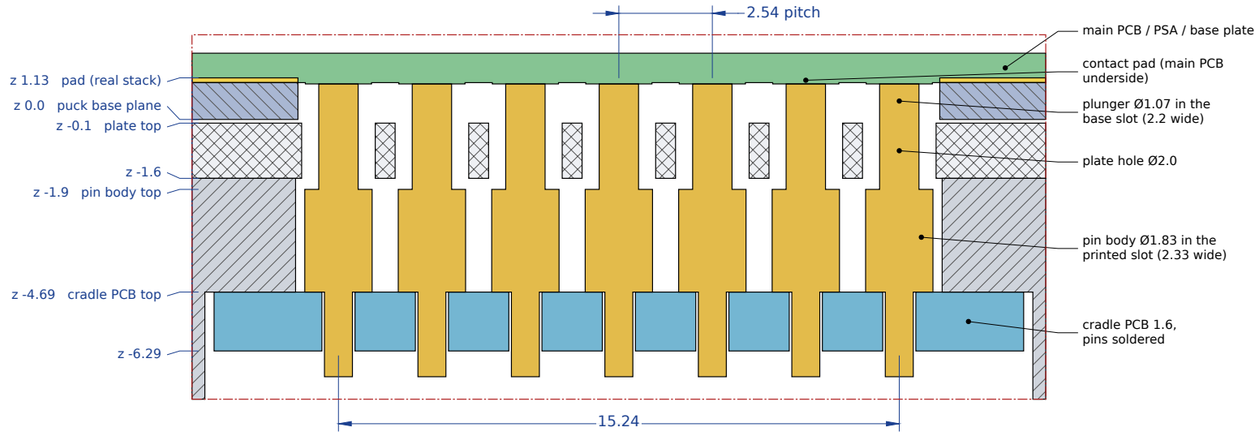
Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.

| | | |
|---|---|---|
| TITLE: **GL1-020  Cradle (charging / data / flip-calibration dock)** | | |
| PROJECT / PART: GreenLine GL-1 electronic ball marker  -  cradle body | | |
| MATERIAL / FINISH: PA12 (MJF) or SLA resin, any colour | PROCESS: MJF or SLA; drill 4 x Ø1.6 after printing | |
| GENERAL TOLERANCE: printed: +-0.3 or +-0.3 % (bureau MJF), as printed | SCALE (A3) / UNITS: 2:1 / 2.8:1  mm | |
| STATUS: **PROVISIONAL - not released** | DWG REV / PARAMS: A-EVT1 / B-EVT1 | SHEET / DATE: 1 of 2  2026-11-11 |

## REFERENCE PLATE GL1-021  2:1  (profile: dxf/GL1-021_cradle_plate.dxf)



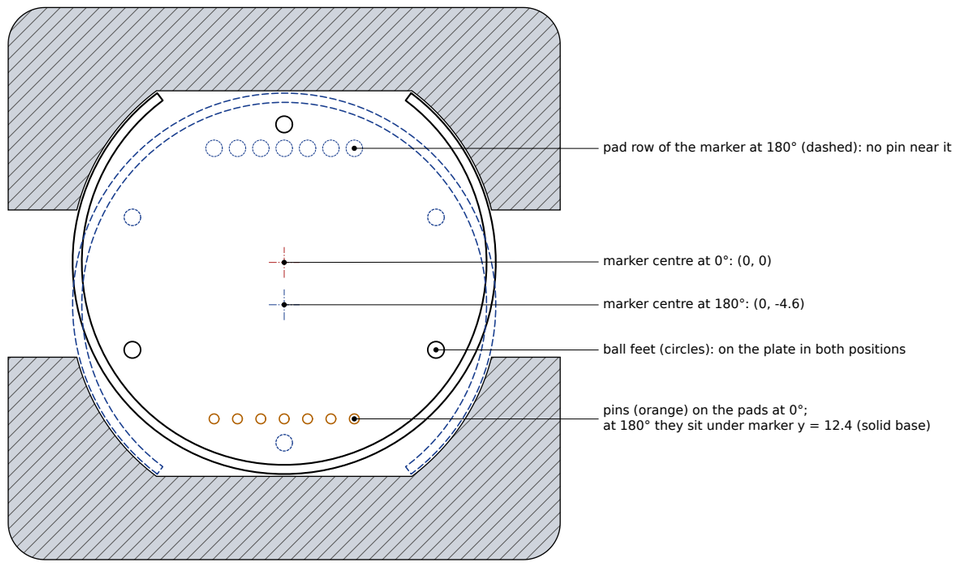

## SECTION E-E  (y = -17.0) through the pin row, marker seated at 0°  6:1




| Cradle PCB (30 x 22 x 1.6, 2 layers) | Value |
|---|---|
| Outline | 22.0 x 30.0 x 1.6, rear edge y -30.5 |
| Pin holes (7) | on y = -17.0, x = -7.62 ... 7.62, pitch 2.54 |
| Screws (4 x M2 thread-forming) | at (+-8.5, -28.0) and (+-8.5, -3.0) |
| USB-C receptacle | GCT USB4105 on the top face, mouth at y -31.8 (rear face recessed 0.5) |
| Top face height | z = -4.69 (pad height 1.13 - working height 5.82) |

Pins: Mill-Max 0914 long stroke (overall 8.3, working height 5.82, 60 gf): diameters PROVISIONAL.
Pin bodies stop 0.3 below the plate so that only the plunger passes the steel (no short through the plate).
The modelled plunger ends on the YAML pad plane (z 0.965); the real pad is at 1.13 with the PSA sheet (D10).
At 180° the plungers bear on the base plate and are compressed 2.27 of 2.29 mm stroke: see README, open issues.

## POSITIONS  1.5:1  marker at 0° (solid) and at 180° (dashed) in the nest




Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.

TITLE
**GL1-020  Cradle (charging / data / flip-calibration dock)**

PROJECT / PART
GreenLine GL-1 electronic ball marker  -  reference plate GL1-021, pin row, positions

| MATERIAL / FINISH | PROCESS | |
|---|---|---|
| plate: 304 stainless 1.5 mm, laser cut, deburred | see sheet 1 | |
| GENERAL TOLERANCE | SCALE (A3) / UNITS | |
| plate: +-0.1, flat within 0.05 | 2:1 / 6:1 / 1.5:1  mm | |
| STATUS | DWG REV / PARAMS | SHEET / DATE |
| **PROVISIONAL - not released** | A-EVT1 / B-EVT1 | 2 of 2  2026-11-11 |